Lecture 9: New Keynesian Policy Design

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Efficient allocation and distortions

Monopolistic competition + flexible prices ⇒ efficient allocation, with a subsidy to correct distortions of monopolistic competition.

Sticky prices + policy that fully stabilizes the price level ⇒ efficient allocation.

Efficient allocation under social planner's problem

$$\max_{\{C_t, N_t\}} U(C_t, N_t)$$

$$\text{s.t.} \qquad C_t(i) = A_t F(N_t(i)), \quad \text{for all } i \in [0,1],$$

$$C_t = \left(\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} \mathrm{d}i\right)^{\frac{\varepsilon}{\varepsilon-1}}, \qquad N_t = \int_0^1 N_t(i)\mathrm{d}i.$$

$$C_t(i) = C_t, \qquad \forall i \in [0,1], \tag{1}$$

$$N_t(i) = N_t, \qquad \forall i \in [0,1], \tag{2}$$

$$-\frac{U_{N,t}}{U_{C,t}} = MPN_t = (1-\alpha)A_t N_t^{-\alpha}, \tag{3}$$

nominal rigidity ⇒ $\begin{cases} C_t(i) \neq C_t, \\ N_t(i) \neq N_t, \end{cases}$

nominal rigidity or monopolistic competition ⇒ (3) does not hold.

(1)&(2) symmetric allocation ⇒ $MPN_t = MPN_t(i)$.

Optimality conditions can be met due to:

i. All goods enter utility symmetrically.

For planner, weight are the same even distribution $C_t(i)$.

ii. Identical technology of production.

For planner, technology for i is the same, even distribution $N_t(i)$.

iii. Concavity of utility.

For planner, there is an optimal allocation of $C_t(i)$ and $N_t(i)$.

$$\mathrm{MRS}_{C,N} = -\frac{U_{N,t}}{U_{C,t}}, \qquad \mathrm{MRT}_{C,N} = \frac{\mathrm{d}C_t}{\mathrm{d}N_t} = F_N(N_t)A_t \quad \text{output transfered one-for-one to consumption.}$$

$$\Rightarrow \mathrm{MRS}_{C,N} = \mathrm{MRT}_{C,N} = \mathrm{MPN}.$$

Suboptimality in NK

Two sources of distortion:

1) Monopolistic competitive firms $\Rightarrow$ market power.

2) Sticky price.

Let's study (1) separately. Assume flexible price. With isoelastic demand function, optimal pricing:

$$P_t = \mu \frac{W_t}{\mathrm{MPN}_t}, \qquad \mu = \frac{\varepsilon}{\varepsilon - 1} \quad \text{markup.}$$

$$\Rightarrow \mathrm{MRS}_{C,N} = -\frac{U_{N,t}}{U_{C,t}} = \frac{W_t}{P_t} = \frac{\mathrm{MPN}_t}{\mu}.$$

$\mu > 1 \Rightarrow$ low employment and output, and optimal condition (3) is violated.

Let τ denotes the employment subsidy; subsidy financed by lump-sum tax, no distortions.

$$P_t = \mu \frac{(1-\tau)W_t}{\mathrm{MPN}_t} \Rightarrow -\frac{U_{N,t}}{U_{C,t}} = \frac{W_t}{P_t} = \frac{\mathrm{MPN}_t}{\mu(1-\tau)}.$$

$$\mathrm{MPN}_t = \frac{\mathrm{MPN}_t}{\mu(1-\tau)} \Rightarrow \mu(1-\tau) = 1 \Rightarrow \tau = \frac{1}{\varepsilon}.$$

The optimal allocation can be achieved by giving employment subsidy of $\tau = \dfrac{1}{\varepsilon}$.

Now study (2) price stickiness.

1° Average markup varies over time

$$\mu_t = \frac{P_t}{(1-\tau)W_t/\mathrm{MPN}_t} = \frac{\mu P_t}{W_t/\mathrm{MPN}_t} \Rightarrow \frac{W_t}{P_t} = \frac{\mu}{\mu_t}\mathrm{MPN}_t.$$

$$-\frac{U_{N,t}}{U_{C,t}} = \frac{W_t}{P_t} = \frac{\mu}{\mu_t}\mathrm{MPN}_t.$$

If $\mu_t \neq \mu$, optimal allocation condition (3) is violated.

2° Staggered price setting: sticky price

$$P_t(i) \neq P_t(j), \qquad C_t(i) \neq C_t(j), \qquad N_t(i) \neq N_t(j).$$

Optimal conditions (1) and (2) are violated.

Optimal Monetary Policy in the basic NK

Assumptions:

1. An optimal subsidy to offset distortion of firms' market power.
2. Economy starts without relative price distortion:

$$P_{-1}(i) = P_{-1}, \qquad \text{for all } i \in [0,1].$$

Optimal policy should achieve:

1. $P_t = P_{t-1}$, $\forall t = 0, 1, 2, \ldots$, so that conditions (1) and (2) hold.
2. $\mu_t = \mu$, $\forall t$, so that condition (3) holds.

$$\Leftrightarrow \tilde{y}_t = \hat{y}_t - \hat{y}_t^n = 0, \quad \forall t; \qquad \hat{\pi}_t = 0, \quad \forall t.$$

DIS:

$$\tilde{y}_t = -\frac{1}{\sigma}\left(\hat{i}_t - \mathbb{E}_t\{\hat{\pi}_{t+1}\} - \hat{r}_t^n\right) + \mathbb{E}_t\{\tilde{y}_{t+1}\}.$$

$$\Rightarrow \hat{i}_t = \hat{r}_t^n, \qquad \forall t.$$

1. Stabilizing output: Not freezing output at constant.

$$\hat{y}_t = \hat{y}_t^n, \qquad \forall t.$$

Eliminating gap, not eliminating fluctuations.

$\hat{y}_t^n$ is subject to technology shock.

2. Stabilizing price: divine coincidence.

If central bank achieves price stability, then it automatically gets $\tilde{y}_t = 0$.

Stabilizing inflation $\Rightarrow$ stabilizing output gap.

How to implement the optimal policy?

Non-policy block:

$$\text{DIS:} \quad \tilde{y}_t = -\frac{1}{\sigma}\left(\hat{i}_t - \mathbb{E}_t\{\hat{\pi}_{t+1}\} - \hat{r}_t^n\right) + \mathbb{E}_t\{\tilde{y}_{t+1}\},$$

$$\text{NKPC:} \quad \hat{\pi}_t = \beta\mathbb{E}_t\{\hat{\pi}_{t+1}\} + \kappa\tilde{y}_t.$$

When $\hat{\pi}_t = 0, \ \forall t \Rightarrow \tilde{y}_t = 0, \ \forall t.$ divine coincidence.

Policy block:

Case a: An exogenous interest rate rule

$$\hat{i}_t = \hat{r}_t^n, \qquad \forall t.$$

$$\Rightarrow \begin{cases} \boxed{\tilde{y}_t} = \dfrac{1}{\sigma}E_t\{\hat{\pi}_{t+1}\} + E_t\{\tilde{y}_{t+1}\}, \\ \hat{\pi}_t = \beta E_t\{\hat{\pi}_{t+1}\} + \kappa\tilde{y}_t \end{cases}$$

Substitute the first equation into the second:

$$\begin{aligned} \hat{\pi}_t &= \beta E_t\{\hat{\pi}_{t+1}\} + \kappa\left(\frac{1}{\sigma}E_t\{\hat{\pi}_{t+1}\} + E_t\{\tilde{y}_{t+1}\}\right) \\ &= \kappa E_t\{\tilde{y}_{t+1}\} + \left(\beta + \frac{\kappa}{\sigma}\right)E_t\{\hat{\pi}_{t+1}\}. \end{aligned}$$

$$\Rightarrow \begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \end{bmatrix} = A_0 \begin{bmatrix} E_t\{\tilde{y}_{t+1}\} \\ E_t\{\hat{\pi}_{t+1}\} \end{bmatrix}, \qquad \text{where } A_0 = \begin{bmatrix} 1 & \frac{1}{\sigma} \\ \kappa & \beta + \frac{\kappa}{\sigma} \end{bmatrix}.$$

$\tilde{y}_t = \hat{\pi}_t = 0$ is one solution associated with optimal policy, but not unique.

$\tilde{y}_t$ and $\hat{\pi}_t$ are neither predetermined.

$$\det(A_0) = \beta, \qquad \operatorname{tr}(A_0) = 1 + \beta + \frac{\kappa}{\sigma}.$$

$$P(1) = 1 - T + D = 1 - 1 - \beta - \frac{\kappa}{\sigma} + \beta = -\frac{\kappa}{\sigma} < 0.$$

$$P(-1) = 1 + T + D = 1 + 1 + \beta + \frac{\kappa}{\sigma} + \beta = 2 + 2\beta + \frac{\kappa}{\sigma} > 0.$$

There is one root between $(-1, 1)$ and another one outside the unit circle.

jump variables $>$ # unstable roots $\Rightarrow$ system is indeterminate.

No guarantee on the realization of $\tilde{y}_t = \hat{\pi}_t = 0, \quad \forall t.$

Case b: A Taylor-type interest rate rule

$$\hat{i}_t = \hat{r}_t^n + \phi_\pi\hat{\pi}_t + \phi_y\tilde{y}_t, \qquad \phi_\pi \geq 0, \quad \phi_y \geq 0, \quad \forall t.$$

$$\begin{cases} \tilde{y}_t = -\frac{\phi_\pi}{\sigma}\hat{\pi}_t - \frac{\phi_y}{\sigma}\tilde{y}_t + \frac{1}{\sigma}\mathbb{E}_t\{\hat{\pi}_{t+1}\} + \mathbb{E}_t\{\tilde{y}_{t+1}\}, \\ \hat{\pi}_t = \beta\mathbb{E}_t\{\hat{\pi}_{t+1}\} + \kappa\tilde{y}_t. \end{cases}$$

$$\begin{cases} \tilde{y}_t = \frac{1-\beta\phi_\pi}{\sigma+\phi_y+\kappa\phi_\pi}\mathbb{E}_t\{\hat{\pi}_{t+1}\} + \frac{\sigma}{\sigma+\phi_y+\kappa\phi_\pi}\mathbb{E}_t\{\tilde{y}_{t+1}\}, \\ \hat{\pi}_t = \frac{\kappa+\beta\sigma+\beta\phi_y}{\sigma+\phi_y+\kappa\phi_\pi}\mathbb{E}_t\{\hat{\pi}_{t+1}\} + \frac{\sigma\kappa}{\sigma+\phi_y+\kappa\phi_\pi}\mathbb{E}_t\{\tilde{y}_{t+1}\}. \end{cases}$$

$$\begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \end{bmatrix} = A_T \begin{bmatrix} \mathbb{E}_t\{\tilde{y}_{t+1}\} \\ \mathbb{E}_t\{\hat{\pi}_{t+1}\} \end{bmatrix},$$

$$A_T = \Omega \begin{bmatrix} \sigma & 1-\beta\phi_\pi \\ \kappa\sigma & \kappa+\beta(\sigma+\phi_y) \end{bmatrix}, \qquad \Omega = \frac{1}{\sigma+\phi_y+\kappa\phi_\pi}.$$

$$\tilde{y}_t = \hat{\pi}_t = 0, \quad \forall t \quad \text{is always a solution.}$$

To guarantee uniqueness, we need two stable roots

$$\text{notice that } x_t = A_T E_t x_{t+1}, \quad \text{but BK is defined on } E_t x_{t+1} = A_T^{-1} x_t,$$

$$2 \text{ stable roots in } A_T \iff 2 \text{ unstable roots in } A_T^{-1}.$$

$$\mathrm{tr}(A_T) = \Omega[\sigma+\kappa+\beta(\sigma+\phi_y)], \qquad \det(A_T) = \Omega\beta\sigma.$$

$$\text{To have stable result, need } D < 1, \quad P(1) > 0, \quad P(-1) > 0.$$

$$\begin{aligned} P(1) = 1 - \mathrm{tr}(A_T) + \det(A_T) &= 1 - \Omega[\sigma+\kappa+\beta\phi_y] > 0 \\ &\Rightarrow \sigma+\kappa+\beta\phi_y < \sigma+\phi_y+\kappa\phi_\pi \\ &\Rightarrow \kappa(\phi_\pi-1) + (1-\beta)\phi_y > 0 \qquad (*) \end{aligned}$$

$$P(-1) = 1 + \mathrm{tr}(A_T) + \det(A_T) = 1 + \Omega[\sigma+\kappa+\beta\phi_y+\beta\sigma] > 0 \quad \text{for sure.}$$

$$\Delta = \mathrm{Tr}(A_T)^2 - 4\det(A_T) = \Omega^2[\sigma+\kappa+\beta(\sigma+\phi_y)]^2 - 4\Omega\beta\sigma, \qquad \text{sign does not matter.}$$

$$\begin{aligned} D = \det(A_T) = \Omega\beta\sigma &= \frac{\beta\sigma}{\sigma+\phi_y+\kappa\phi_\pi} = \frac{1}{\frac{1}{\beta}+\frac{\phi_y+\kappa\phi_\pi}{\beta\sigma}} \\ &< \frac{1}{1+\frac{\kappa-\kappa\phi_\pi+\kappa\phi_\pi}{\beta\sigma}} = \frac{1}{1+\frac{\kappa}{\beta\sigma}} < 1. \end{aligned}$$

$$\text{If } \kappa(\phi_\pi-1)+(1-\beta)\phi_y > 0 \text{ is satisfied}, \quad \tilde{y}_t = \hat{\pi}_t = 0 \quad \text{and} \quad \hat{i}_t = \hat{r}^n_t \text{ hold} \forall t.$$

$$\text{If } \phi_y = 0 \Rightarrow \kappa(\phi_\pi-1) > 0 \Rightarrow \phi_\pi > 1 \quad \text{Taylor principle.}$$

If $\phi_y > 0$, output gap helps, lower bound on ϕ_π can be relaxed a little.

But $1-\beta$ is usually small, inflation is the main stabilizing force.

Case c: A forward-looking interest rate rule

$$\hat{i}_t = \hat{r}_t^n + \phi_\pi \mathbb{E}_t\{\hat{\pi}_{t+1}\} + \phi_y \mathbb{E}_t\{\tilde{y}_{t+1}\}.$$

$$\begin{cases} \tilde{y}_t = \dfrac{1-\phi_\pi}{\sigma}\mathbb{E}_t\{\hat{\pi}_{t+1}\} + \dfrac{\sigma-\phi_y}{\sigma}\mathbb{E}_t\{\tilde{y}_{t+1}\}, \\ \hat{\pi}_t = \beta \mathbb{E}_t\{\hat{\pi}_{t+1}\} + \kappa \tilde{y}_t. \qquad (*) \end{cases}$$

$$\begin{cases} \tilde{y}_t = \dfrac{1-\phi_\pi}{\sigma}\mathbb{E}_t\{\hat{\pi}_{t+1}\} + \dfrac{\sigma-\phi_y}{\sigma}\mathbb{E}_t\{\tilde{y}_{t+1}\}, \qquad (**) \\ \hat{\pi}_t = \dfrac{\kappa(1-\phi_\pi)+\beta\sigma}{\sigma}\mathbb{E}_t\{\hat{\pi}_{t+1}\} + \dfrac{\kappa(\sigma-\phi_y)}{\sigma}\mathbb{E}_t\{\tilde{y}_{t+1}\}. \qquad (***) \end{cases}$$

$$\begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \end{bmatrix} = A_F \begin{bmatrix} \mathbb{E}_t\{\tilde{y}_{t+1}\} \\ \mathbb{E}_t\{\hat{\pi}_{t+1}\} \end{bmatrix},$$

$$A_F = \begin{bmatrix} \frac{\sigma-\phi_y}{\sigma} & \frac{1-\phi_\pi}{\sigma} \\ \frac{\kappa(\sigma-\phi_y)}{\sigma} & \beta + \frac{\kappa(1-\phi_\pi)}{\sigma} \end{bmatrix}.$$

$$\operatorname{tr}(A_F) = \beta + \frac{\kappa(1-\phi_\pi)+\sigma-\phi_y}{\sigma}, \qquad \det(A_F) = \beta\frac{\sigma-\phi_y}{\sigma}.$$

$$\begin{aligned} P(1) &= 1 - \operatorname{tr}(A_F) + \det(A_F) \\ &= 1 - \beta - \frac{\kappa(1-\phi_\pi)+(\sigma-\phi_y)}{\sigma} + \beta\frac{\sigma-\phi_y}{\sigma} > 0 \\ &\Rightarrow \kappa(\phi_\pi - 1) + (1-\beta)\phi_y > 0 \qquad (1) \quad \text{policy cannot be too passive.} \end{aligned}$$

$$\begin{aligned} P(-1) &= 1 + \operatorname{tr}(A_F) + \det(A_F) \\ &= 1 + \beta + \frac{\kappa}{\sigma}(1-\phi_\pi) + 1 - \frac{1}{\sigma}\phi_y + \beta - \frac{\beta}{\sigma}\phi_y > 0 \\ &\Rightarrow \kappa(\phi_\pi - 1) + (1+\beta)\phi_y < 2\sigma(1+\beta) \qquad (2) \quad \text{policy cannot be too aggressive.} \end{aligned}$$

$$D < 1 \Rightarrow \beta\left(1 - \frac{\phi_y}{\sigma}\right) < 1, \quad \text{since } \beta \in (0,1) \text{ and } \phi_y \geq 0, \text{ so this holds.}$$

$$\Rightarrow \text{Under (1) and (2), we have 2 stable roots in } A_F, \text{ 2 unstable roots in } A_F^{-1}.$$

$$\tilde{y}_t = \hat{\pi}_t = 0, \quad \forall t, \quad \text{is the unique solution.}$$

1) If $\phi_\pi > 1$, the coefficient on $\mathbb{E}_t\{\hat{\pi}_{t+1}\}$ is negative.

$$\mathbb{E}_t\{\hat{\pi}_{t+1}\} \uparrow \Rightarrow \tilde{y}_t \downarrow \Rightarrow \hat{\pi}_t \downarrow.$$

If $\phi_\pi < 1$, then

$$\mathbb{E}_t\{\hat{\pi}_{t+1}\} \uparrow \Rightarrow \tilde{y}_t \uparrow \Rightarrow \hat{\pi}_t \uparrow,$$

self-fulfilling inflation; policy cannot be too passive.

2) If policy is too aggressive so that condition (2) is violated,

$$P(-1) < 0, \quad \text{one eigenvalue crosses } -1, \quad \text{system unstable.}$$

But all optimal policy listed are not realistic, since policy rate must be adjusted one-for-one with the natural interest rate.

But $\hat{r}_t^n$ is usually unobservable, because it depends on the true model of the economy,

parameter values and the realized shocks. $\Rightarrow$ Simple rules are needed!

Due to divine coincidence, unobservable output gap is not the main issue, since $E_t\hat{\pi}_{t+1} = 0 \Rightarrow \tilde{y}_t = 0$.

$$\kappa(\phi_\pi - 1) + (1-\beta)\phi_y = 0 \quad \Longleftrightarrow \quad \phi_\pi = 1 - \frac{1-\beta}{\kappa}\phi_y.$$

$$\kappa(\phi_\pi - 1) + (1+\beta)\phi_y = 2\sigma(1+\beta) \quad \Longleftrightarrow \quad \phi_\pi = 1 + \frac{2\sigma(1+\beta)}{\kappa} - \frac{1+\beta}{\kappa}\phi_y.$$

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图 1: Determinacy region

Simple monetary policy rule

1. Rotemberg and Woodford (1999)

Second-order approximation to the utility losses due to deviations from efficient allocation:

$$W = \frac{1}{2}\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\left(\sigma + \frac{\varphi+\alpha}{1-\alpha} \right) \tilde{y}_t^2 + \frac{\varepsilon}{\kappa}\hat{\pi}_t^2 \right].$$

Per-period loss:

$$L = \frac{1}{2}\left[\left(\sigma + \frac{\varphi+\alpha}{1-\alpha} \right) \mathrm{Var}(\tilde{y}_t) + \frac{\varepsilon}{\kappa}\mathrm{Var}(\hat{\pi}_t) \right].$$

$\sigma\uparrow,\ \varphi\uparrow,\ \alpha\uparrow \Rightarrow L\uparrow$. Curvature amplifies the effect (consumption, labor, technology).

$\varepsilon\uparrow,\ \frac{1}{\kappa}\uparrow \Rightarrow L\uparrow$. Amplify losses from price dispersion; $\frac{1}{\lambda}$ amplifies the degree of price dispersion.

2. Taylor-type interest rate rule

$$\hat{i}_t = \phi_\pi \hat{\pi}_t + \phi_y \hat{y}_t, \qquad \phi_\pi > 0, \quad \phi_y > 0.$$
$$\hat{i}_t = \phi_\pi \hat{\pi}_t + \phi_y \tilde{y}_t + \phi_y \hat{y}_t^n.$$

Plug in:

$$\begin{cases} \text{DIS:} \quad (\sigma + \phi_y)\tilde{y}_t + \phi_\pi \hat{\pi}_t = \sigma \mathbb{E}_t\{\tilde{y}_{t+1}\} + \mathbb{E}_t\{\hat{\pi}_{t+1}\} + (\hat{r}_t^n - \phi_y \hat{y}_t^n), \\ \qquad\qquad \text{NKPC:} \quad \hat{\pi}_t = \beta \mathbb{E}_t\{\hat{\pi}_{t+1}\} + \kappa \tilde{y}_t. \end{cases}$$

$$\begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \end{bmatrix} = A_T \begin{bmatrix} \mathbb{E}_t\{\tilde{y}_{t+1}\} \\ \mathbb{E}_t\{\hat{\pi}_{t+1}\} \end{bmatrix} + B_T(\hat{r}_t^n - \phi_y \hat{y}_t^n),$$

$$A_T = \Omega \begin{bmatrix} \sigma & 1 - \beta\phi_\pi \\ \kappa\sigma & \kappa + \beta(\sigma + \phi_y) \end{bmatrix}, \quad B_T = \Omega \begin{bmatrix} 1 \\ \kappa \end{bmatrix}, \quad \Omega = \frac{1}{\sigma + \phi_y + \kappa\phi_\pi}.$$

Assume

$$\hat{a}_t = \rho_a \hat{a}_{t-1} + \varepsilon_t^a, \qquad \hat{y}_t^n = \psi_{ya}^n \hat{a}_t, \qquad \psi_{ya}^n = \frac{1+\phi}{\sigma + \phi + \alpha(1-\sigma)} > 0.$$

$$\Rightarrow v_t = \phi_y \hat{y}_t^n = \phi_y \psi_{ya}^n \hat{a}_t.$$

$$\hat{r}_t^n = \sigma \mathbb{E}_t\{\Delta \hat{y}_{t+1}^n\} = -\sigma \psi_{ya}^n (1-\rho_a)\hat{a}_t.$$

$$\Delta \hat{y}_{t+1}^n = \hat{y}_{t+1}^n - \hat{y}_t^n = \psi_{ya}^n(\hat{a}_{t+1} - \hat{a}_t), \qquad \mathbb{E}_t \Delta \hat{y}_{t+1}^n = \psi_{ya}^n(\rho_a - 1)\hat{a}_t.$$

$$\hat{r}_t^n - v_t = -\psi_{ya}^n[\sigma(1-\rho_a) + \phi_y]\hat{a}_t.$$

$$B_T(\hat{r}_t^n - v_t) = \begin{bmatrix} 1 \\ \kappa \end{bmatrix} \Omega \left\{ -\psi_{ya}^n \left[\sigma(1-\rho_a) + \phi_y\right] \right\} \hat{a}_t.$$

Shock loading: $\qquad f(\phi_y) = \dfrac{\sigma(1-\rho_a) + \phi_y}{\sigma + \phi_y + \kappa\phi_\pi}, \qquad f'(\phi_y) = \dfrac{\sigma\rho_a + \kappa\phi_\pi}{(\sigma + \phi_y + \kappa\phi_\pi)^2} > 0.$

Shock term gets larger if the central bank responds more aggressively to output.

$\Rightarrow$ output gap and inflation variance $\uparrow$, welfare $\downarrow$.

$\phi_\pi \to +\infty \Rightarrow f(\phi_y) \to 0 \Rightarrow$ the exogenous driving force hitting the system goes to 0.

$\Rightarrow$ Aggressive inflation targeting gets arbitrarily close to the optimum.

A constant money growth rule: Friedman (1960)

$$\Delta \hat{m}_t = 0, \qquad \forall t.$$

$$\hat{m}_t - \hat{p}_t = \hat{y}_t - \eta \hat{i}_t - \xi_t, \qquad \xi_t \text{ is an exogenous money demand shock: } \Delta \xi_t = \rho_\xi \Delta \xi_{t-1} + \varepsilon_t^\xi.$$

Let

$$\widehat{\mathbb{M}}_t^+ = \widehat{\mathbb{M}}_t + \xi_t, \qquad \text{denotes real balance adjusted by exogenous component of money demand.}$$

$$\widehat{\mathbb{M}}_t^+ = \tilde{y}_t + \hat{y}_t^n - \eta \hat{i}_t.$$

$$\Rightarrow \hat{i}_t = \frac{1}{\eta}\left(\tilde{y}_t + \hat{y}_t^n - \widehat{\mathbb{M}}_t^+\right).$$

$$\widehat{\mathbb{M}}_t^+ - \widehat{\mathbb{M}}_{t-1}^+ = \hat{m}_t - \hat{m}_{t-1} - \hat{p}_t + \hat{p}_{t-1} + \xi_t - \xi_{t-1} = 0 - \hat{\pi}_t + \Delta \xi_t.$$

$$\Rightarrow \widehat{\mathbb{M}}_{t-1}^+ = \widehat{\mathbb{M}}_t^+ + \hat{\pi}_t - \Delta \xi_t.$$

$$A_{M,0} \begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \\ \widehat{\mathbb{M}}_{t-1}^+ \end{bmatrix} = A_{M,1} \begin{bmatrix} \mathbb{E}_t\{\tilde{y}_{t+1}\} \\ \mathbb{E}_t\{\hat{\pi}_{t+1}\} \\ \widehat{\mathbb{M}}_t^+ \end{bmatrix} + B_M \begin{bmatrix} \hat{r}_t^n \\ \hat{y}_t^n \\ \Delta \xi_t \end{bmatrix}.$$

The degree of stability of money demand is a key element in determining the desirability of a money growth rule.